

## Blastocystis subtypes isolated from irritable bowel syndrome patients and co-infection with *Helicobacter pylori*.

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Irritable bowel syndrome (IBS) is a chronic functional gastrointestinal disease presenting clinically by abdominal pain with alteration of bowel habits. Although IBS has uncertain etiology, chronic gut inflammation due to persistent exposure to an infectious agent including *Blastocystis* sp. was proposed. The aim of this study was to detect the prevalence of *Blastocystis* sp. subtype (ST) isolated from stool of IBS patients and to assess *Blastocystis* sp. and *H. pylori* co-infection in IBS patients from Beni-Suef Governorate, Egypt. Stool samples were collected from 115 IBS patients, following Rome III criteria. All stool samples were microscopically examined by wet mount and permanent trichrome stain, cultured on Jones' medium with further sequencing of positive *Blastocystis* isolates and screened for detection of *H. pylori* coproantigen. *Blastocystis* sp. was the predominant parasite in IBS patients; it had statistical significant association with both rural residence (OR = 10) and flatulence (OR = 8.2). There was a predominance of *Blastocystis* sp. ST3 (81%) followed by ST1 (19%). *Blastocystis* culture results (19.1%) were superior than microscopy (16.5%). The majority of *Blastocystis*-positive IBS patients (72.7%) were co-infected with *H. pylori* with statistical significance; however, *H. pylori* was higher in *Blastocystis*-negative IBS patients (47/64) than in *Blastocystis*-positive IBS patients (17/64). Interestingly, IBS is usually associated with gut dysbiosis, while the most prevalent parasite in our IBS patients was *Blastocystis* sp., which is frequently found in asymptomatic individuals. Whether *Blastocystis* sp. is a cause or a consequence of IBS still needs further investigation, with a particular focus on correlation of IBS with different *Blastocystis* sp. subtypes and gut microbiomes.

## On the fiftieth anniversary. Postinfectious irritable bowel syndrome: mechanisms related to pathogens.

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Gastrointestinal (GI) infections resulting from bacterial, viral, and parasitic pathogens predispose to postinfectious irritable bowel syndrome (PI-IBS) and other functional GI disorders. Existing literature supports the role of enterochromaffin cell hyperplasia, serotonin synthesis and reuptake, impaired barrier function, altered immune activation, and potentially mast cell activation in the pathophysiology of PI-IBS. The objective of this review was to summarize from the literature the characteristics of the pathogens commonly implicated in PI-IBS, their acute enteritis phases, and the changes seen in the postinfectious phase that may contribute toward development of IBS. A limitation of our current understanding is that the postinfectious GI sequelae reported in prior studies followed epidemic diarrheal outbreaks often involving more than one pathogen, or the studies focused on highly selected, tertiary referral patients. Understanding the mechanisms, natural history, and optimized management of individuals suffering PI-IBS following the more typical sporadic infection requires larger studies of PI-IBS following GI infections encountered in community settings. These studies should include genetic, physiological, and molecular studies to provide more generalizable information that can ultimately be used to diagnose, manage, and potentially prevent the development of PI-IBS.

## Postinfectious irritable bowel syndrome.

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Irritable bowel syndrome (IBS) is a gastrointestinal disorder characterized by abdominal pain and changes in bowel habits, not sustained by structural changes. There is now consistent evidence indicating that IBS may be the adverse outcome of an acute episode of infectious gastroenteritis, the so-called postinfectious (PI) IBS. The infectious agents involved in the development of PI-IBS include pathogenic bacteria, parasites, and viruses. Abdominal pain and diarrhea are the most common symptoms of PI-IBS. Several studies identified a number of risk factors increasing the susceptibility for PI-IBS development. These include the virulence of the pathogen, the severity, and duration of the acute enteritis, younger age, female sex, and psychological disturbances. Several mucosal abnormalities in the colon or ileum of patients who develop PI-IBS have been described. These changes include increased mucosal permeability, an increased amount of intraepithelial lymphocytes, lamina propria T cells, and mast cells, as well as serotonin-containing enteroendocrine cells. The mediators released by these activated cells may evoke enteric nervous system responses, excite sensory afferent pathways, and induce visceral hyperalgesia. Little is known about the prognosis of PI-IBS, although it is likely better than that of nonspecific IBS. There is little evidence about a specific treatment for PI-IBS. Although probiotics and antibiotics may be promising in the prevention of PI-IBS, the efficacy of these treatments should be assessed in an ad hoc designed study.

## Salmonella gastroenteritis during childhood is a risk factor for irritable bowel syndrome in adulthood.

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Acute infectious gastroenteritis increases the risk for irritable bowel syndrome (IBS) and functional dyspepsia (FD). Children are particularly vulnerable to gastroenteritis because of the immaturity of their intestinal barrier, enteric nervous system, and immune response to pathogens. We investigated whether acute gastroenteritis in early life increases the risk of IBS and FD throughout adulthood. In 1994, we identified and monitored a single culture-proven foodborne *Salmonella enteritidis* outbreak that involved 1811 patients (mostly pediatric) in Bologna, Italy. Clinical data were collected and a prospective, controlled, cohort study was designed. Long-term effects were assessed by mailing a questionnaire to 757 subjects 16 years after the outbreak (when all of the children were adults). We randomly selected a cohort of 250 adults exposed to *Salmonella* as children, all 127 individuals exposed as adults, and a cohort of nonexposed participants matched for number, age, sex, and area of residence (controls). Among 198 exposed participants, 64 reported FD (32.3%), compared with 51 of 188 controls (27.1%;  $P = .268$ ). Among 204 exposed participants, 75 reported having IBS (36.8%) compared with 44 of 189 controls (23.3%;  $P = .004$ ). The odds ratio for IBS among people exposed to the *Salmonella* was 1.92 (95% confidence interval: 1.23-2.98). The prevalence of IBS was higher in individuals exposed *Salmonella* as children than in controls (35.3% vs 20.5%;  $P = .008$ ), but not in individuals exposed as adults, compared with controls. After multivariate logistic regression, post-infectious IBS was independently associated with anxiety and FD. Based on data collected from a single culture-proven foodborne *Salmonella enteritidis* outbreak in 1994, *Salmonella*-induced gastroenteritis during childhood (but not adulthood) is a risk factor for IBS.

[Post-infectious irritable bowel syndrome. A review based on current evidence].

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Pathophysiology of irritable bowel syndrome (IBS) is multifactorial. Recent investigations have associated episodes of infectious gastroenteritis with development of IBS. This condition is named post-infectious IBS (PI-IBS). The role of inflammation-infection in IBS pathogenesis is not well understood. To review published scientific evidence on PI-IBS regarding risk factors, causal agents, histopathological changes, and treatment. An electronic search in MEDLINE and abstracts presented at national and international GI meetings was performed, looking for information published in the past 50 years including animal studies, cohort studies, case-control studies, and series of cases and case reports, using the key words post-infectious enteritis, post-dysenteric or post-infectious irritable bowel syndrome (PI-IBS), and post-infectious colitis. Fifty one papers were included. These studies were classified according to pathophysiologic mechanisms, infectious agents involved, animal or human studies, and treatment. Current evidence shows a strong association between colonic infection and inflammation with development of IBS. Approximately 25% of patients with IBS have a history of infectious enteritis. Microbial agents related with PI-IBS include bacteria (*Campylobacter*, *Salmonella*) and parasites (*Trichinella spiralis*). Increased number of enteroendocrine cells, CD3 lymphocytes and mast cells within the colonic muscle wall, release of pro-inflammatory substances, and increased number of inflammatory cells with intestinal nervous endings are the most common histopathologic findings. Patients developing PI-IBS have a higher frequency of psychological disorders and stressful events prior to the gastroenteritis episode. Therapeutic interventions with steroids, COX-2 inhibitors, antibiotics and probiotics require further investigation.

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## Potential Causes and Present Pharmacotherapy of Irritable Bowel Syndrome: An Overview.

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Irritable bowel syndrome (IBS) is currently one of the most common disorders of the digestive system in the Western society. Almost 2 out of 10 people suffer from IBS with women being more affected than men. IBS is associated with abdominal pain, bloating and altered stool consistency and imposes a heavy burden for the affected patients. The pathophysiology of IBS remains elusive although potential causes have been suggested, such as a deranged brain-gut signaling, hypersensitivity of visceral sensory afferent fibers, bacterial gastroenteritis, small intestinal bacterial overgrowth (SIBO), genetic alterations and food sensitivity. Targets for the pharmacotherapy of IBS include the serotonergic and opioidergic system, and the microbial population of the gut. Alternative therapies like traditional Chinese medicine have shown some success in the combat against IBS. Key Messages: Many therapeutics for the treatment of IBS have emerged in the past; however, only a few have met up with the expectations in larger clinical trials. Additionally, the multifactorial etiology of IBS and its variety of cardinal symptoms requires an individual set of therapeutics. This review provides a short overview of potential causes and current pharmacological therapeutics and of additional and alternative therapies for IBS.

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## Irritable bowel syndrome: a review on the role of intestinal protozoa and the importance of their detection and diagnosis.

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Irritable bowel syndrome (IBS) is a functional gastrointestinal disorder in which abdominal pain is associated with a defect or a change in bowel habits. Gut inflammation is one of the proposed mechanisms of pathogenesis. Recent studies have described a possible role for protozoan parasites, such as *Blastocystis hominis* and *Dientamoeba fragilis*, in the etiology of IBS. *Dientamoeba fragilis* is known to cause IBS-like symptoms and has a propensity to cause chronic infections but its diagnosis relies on microscopy of stained smears, which many laboratories do not perform, thereby leading to the misdiagnosis of *dientamoebiasis* as IBS. The role of *B. hominis* as an etiological agent of IBS is inconclusive, due to contradictory reports and the controversial nature of *B. hominis* as a human pathogen. Although *Entamoeba histolytica* infections occur predominately in developing regions of the world, clinical diagnosis of amebiasis is often difficult because symptoms of patients with IBS may closely mimic those patients with non-dysenteric amoebic colitis. Clinical manifestations of *Giardia intestinalis* infection also vary from asymptomatic carriage to acute and chronic diarrhoea with abdominal pain. These IBS-like symptoms can be continuous, intermittent, sporadic or recurrent, sometimes lasting years without correct diagnosis. It is essential that all patients with IBS undergo routine parasitological investigations in order to rule out the presence of protozoan parasites as the causative agents of the clinical signs.

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## Irritable bowel syndrome and intestinal parasites: a view from South America.

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Irritable bowel syndrome (IBS) is a functional gastrointestinal disorder of uncertain etiology. Several studies have proposed the possible role of intestinal parasites in the pathogenesis of IBS. We aimed to summarize the epidemiological studies that describe a possible link between intestinal parasites and IBS, with special interest in endemic areas for intestinal parasitism such as South America. A comprehensive review of the literature was conducted by using the keywords: irritable bowel syndrome, intestinal parasites, protozoan infection, soil-transmitted helminths and South America. *Giardia lamblia* may cause IBS symptoms that can persist several years after effective treatment. *Dientamoeba fragilis* can cause IBS-like symptoms, but low sensitive parasitological techniques may fail to detect it. *Entamoeba histolytica* can cause a chronic non-dysenteric colitis, but several studies have failed to find an association with IBS. The role of *Blastocystis hominis* in IBS remains controversial. In addition, epidemiological studies evaluating the effect of soil-transmitted helminths in IBS are scant. Symptoms elicited by intestinal parasites may resemble to those in IBS, especially in endemic areas such as South America, where both the prevalence of IBS and intestinal parasitism are high. Whether these organisms are the cause or contributing factors in IBS remains a subject of study. Routine parasitological examination of stools in individuals who full-fit the criteria for IBS should be included upon initial assessment in endemic countries.

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## Is irritable bowel syndrome an inflammatory disorder?

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Histopathologic data demonstrate low-grade mucosal inflammation in a subset of patients with irritable bowel syndrome (IBS). This inflammatory infiltrate is mainly represented by increased numbers of T lymphocytes and mast cells lying in the lamina propria. The close apposition of immunocytes to gut nerves supplying the mucosa provides a basis for neuroimmune cross-talk, which may explain gut

sensorimotor dysfunction and related symptoms in patients with IBS. A previous gastroenteritis (due to *Campylobacter jejuni*, *Salmonella*, *Shigella*, *Escherichia coli*, and, likely, viruses) is now an established etiologic factor for IBS (hence, postinfectious IBS). Other putative causes, such as undiagnosed food allergies, genetic abnormalities, stress, or bile acid malabsorption, may also promote and maintain a low-grade mucosal inflammation in IBS. The identification of mucosal inflammation in IBS has pathophysiologic implications and paves the way for novel therapeutic options.

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### Irritable bowel syndrome: in search of an etiology: role of *Blastocystis hominis*.

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This study was designed to examine stool specimens of irritable bowel syndrome (IBS) patients for *Blastocystis hominis*, a common intestinal parasite. One hundred fifty patients were enrolled, 95 IBS cases and 55 controls. These patients provided a medical history, and underwent physical and laboratory evaluations that included stool microscopy and culture for *B. hominis* and colonoscopy. The 95 cases (51 males and 44 females) had a mean  $\pm$  SD age of 37.8  $\pm$  13.2 years. Stool microscopy was positive for *B. hominis* in 32% (30 of 95) of the cases and 7% (4 of 55) of the controls ( $P = 0.001$ ). Stool culture was positive in 46% (44 of 95) of the cases and 7% (4 of 55) of the controls ( $P < 0.001$ ). Stool culture for *B. hominis* in IBS was more sensitive than microscopy ( $P < 0.001$ ). *Blastocystis hominis* was frequently demonstrated in the stool samples of IBS patients; however, its significance in IBS still needs to be investigated. Stool culture has a higher positive yield for *B. hominis* than stool microscopy.

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